Activity Ownership & “The User’s Namespace”

Layering in the Social





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vCard | Web | DIDComm | DID

Who dis?

DIF

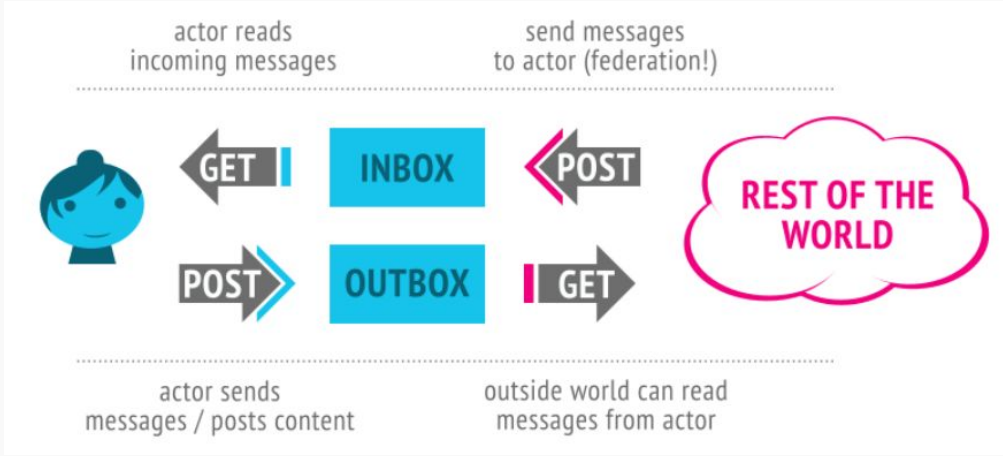
- Represent Decentralized Identity Foundation (now a W3C Member!), in the Linux Fdtn
- Long-time DID and ActivityPub enthusiast
- Work history: identity startups, fintech, blockchain, social networking
- Communities: Internet Identity Workshop, MyData, Chaos Computer Club

Terminological nuances

- End-User vs. Actor
- Social Web vs. Social Media/Network
- Data Ownership vs. Data Control
- Web Server vs. Outbox Server
- Account Portability (identifier changes)
  vs. Data Portability (export, forwarding)
  vs. Service Portability (provider changes)



Inbox/Outbox (SMTP) first, Public Streams (RSS) second



Fundamentally, in AP, an Actor is **not** a publisher, an “account”, or a long-lived public or human-readable URL.

The Actor is a person or process with an inbox and an outbox, period. The Actor’s “URL” is a machine-readable way to find their (current, mutable) inbox and outbox URLs.

Public and high-volume streaming messages are layered on top as a special case. Fundamentally it’s a messaging protocol.

The Hard Parts: Identity, AuthZ

- AP is **just** a “narrow waist” for passing messaging: **no** MUSTs about identity, storage, or authorization, only confusingly vague SHOULDs and MAYs
- Explicitly calls out cross-origin outboxes and services
- Identity, storage, and authorization were expected to be **profiled** for different use-cases and communities
- Mastodon (single-origin) emerged as only major profile… with OAuth, Webfinger, and no particular storage baselines or APIs
- SWICG now hardening this ONE profile



User-empowering software is easier to build now!

For Actors to have service portability and control their identifiers, they need at minimum “cross-origin identifiers” and authorization mechanisms for “cross-origin storage” of some kind

This could mean DIDs, or indie-web/self-hosted CIDs, and/or linked web storage and a compatible storage ecosystem (MyData Operators, or W3C-CCG Portable Web Spaces, or a multi-protocol extension to Atproto PDS)

Identifiers *MUST* be provided for activities posted in server to server communication, unless the activity is intentionally transient. However, for client to server communication, a server receiving an object posted to the outbox with no specified `id` *SHOULD* allocate an object ID in the actor's namespace and attach it to the posted object.

Src: 2017 CR of AP Spec



Call to Action: Profile + Prototype

The ActivityPub work can be profiled many ways, particularly for non-”social media” use cases:

- package managers
- git forges
- “data spaces” and data unions
- agentic inbox/outboxes
- confidential exchange of Verifiable Credentials

But it’s hard to lobby for the profile hinge if there’s no one using anything else in prod!

Social Web CG:

Social Web WG: